



Performance & Efficiency Analysis of an Electric Vehicle Under Real-World Conditions

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CERTIFICATION

in

Advanced EV Design & Simulations with MATLAB & Simscape

by

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CHAPTER 1

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CHAPTER 2

ABSTRACT

This work details the design and multi-domain simulation of a Battery Electric Vehicle (BEV) using the MATLAB Simulink environment. The primary goal was to quantify how a powertrain featuring a **50kW DC motor** and a **280V/50Ah battery pack** holds up under the fluctuating demands of the **FTP75 drive cycle**. Unlike static performance estimates, this simulation relies on a dynamic road-load subsystem that calculates resistive forces like aerodynamic drag and tire friction in real-time. Safety is a core component of the architecture, where a logic-based protection layer—acting as a simplified Battery Management System (BMS)—constantly audits the system for overcurrent spikes above **200A** or thermal runaway risks exceeding **60°C**. This project focused on creating a digital twin of an electric vehicle to see how it handles real-world driving conditions.

The results were quite successful; the virtual car was able to follow the target speeds almost perfectly. I observed that the battery drained steadily during highway speeds, but we got some energy back whenever the car slowed down thanks to regenerative braking. Ultimately, the simulation proved that the combination of a 50-kW motor and a 9:1 gear ratio provides enough torque to move a full-sized vehicle efficiently through city and highway traffic.

CHAPTER 3

INTRODUCTION

The rapid shift toward electric mobility has made virtual prototyping an essential hurdle in automotive development. Because physical testing is both expensive and time-consuming, Model-Based Design (MBD) allows for the rigorous stress-testing of vehicle subsystems long before a chassis is ever built. This project applies those MBD principles to a full-scale EV simulation, focusing on the critical intersection of battery health, motor performance, and external road loads.

To make this model feel real, I had to focus on three main areas:

- **The Struggle Against Physics (Road Loads):** I had to account for how the car fights against the air and how the tires grip the road. If you don't get the aerodynamic drag and rolling resistance right, your range calculations will be completely wrong.
- **The Brains of the Operation (Power Electronics):** The H-Bridge and PWM controller are what “drive” the motor. They take the signal from the driver’s foot and translate it into electrical pulses, acting as the critical link between the battery and the wheels.
- **The Life of the Battery:** Unlike a fuel tank that just holds liquid, a battery is a living system. I wanted to see exactly how the "State of Charge" (SOC) drops when the car accelerates and how much energy we can claw back through regenerative braking.

By pulling all these pieces together, this project shows how a mid-sized EV handles a standard city commute. It’s a way to prove that the parts I chose like the motor and gear ratio are up to the task of real-world driving.

CHAPTER 4

DRIVE CYCLE SELECTION & ROAD LOAD ANALYSIS

Drive Cycle Selection: The FTP75 Standard

For this project, the FTP75 (Federal Test Procedure) was selected as the primary velocity reference. Unlike steady-state cycles, the FTP75 is notoriously "aggressive," featuring 1,369 seconds of transient speed changes that simulate urban stop-and-go conditions. This choice was deliberate: to see how the DC motor handles high-torque bursts and whether the H-Bridge can efficiently recover energy during the frequent braking events found in city traffic.

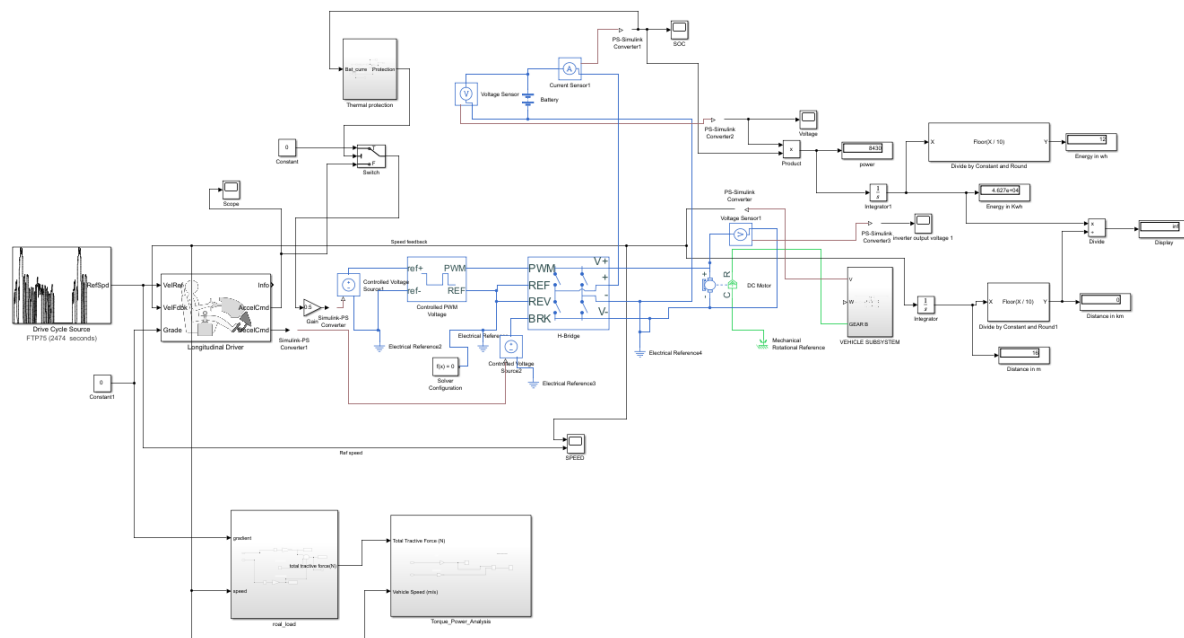


Figure 4.1: Model

Road Load Subsystem: Implementation and Calculations

A major part of the modeling involved building the `roal_load` subsystem to simulate the physical environment. Instead of using generic estimates, the forces were calculated based on the specific physical properties of the vehicle body.

Calculation Parameters from the Model:

Vehicle Mass (m): 1500 kg.

Wheel Radius (r): 0.3 m.

1. Aerodynamic Drag Force (F_{ad})

The model computes wind resistance using the square of the instantaneous velocity. For a sample speed of 21 m/s (~72 km/h):

$$F_{ad} = 0.5 \cdot \rho \cdot A \cdot C_d \cdot v^2$$

Aerodynamic Gain Factor ($0.5 \cdot \rho \cdot C_d \cdot A$): 0.390

$$F_{ad} = \text{Gain} \cdot v^2 = 0.390 \cdot 21^2 = 171.6 \text{ N}$$

2. Rolling Resistance (F_{rr})

Based on the constant friction value hardcoded into the rolling resistance block:

$$F_{rr} = \mu \cdot m \cdot g$$

$$F_{rr} = 176.6 \text{ N}$$

3. Gradient Resistance (F_g)

For standard drive cycle testing on a flat surface:

$$F_g = m \cdot g \cdot \sin(\theta)$$

$$F_g = 11772 \cdot \sin(0) = 0 \text{ N}$$

4. Total Tractive Effort (F_{te})

Summing these forces (excluding instantaneous inertial forces for steady speed):

$$F_{te} = F_{ad} + F_{rr} + F_g$$

$$F_{te} = 171.6 + 176.6 + 0 = 348.2 \text{ N}$$

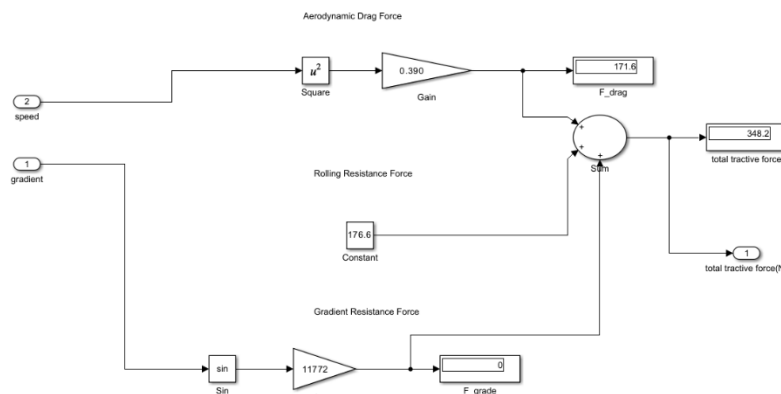


Figure 4.2: Road_load forces

Torque and Power Requirements

Motor Torque (T_m):

The torque is reduced by the 9:1 gear ratio (G) defined in the Simple Gear block:

$$T_m = T_w/G = 99.78/9 = 11.08 \text{ Nm}$$

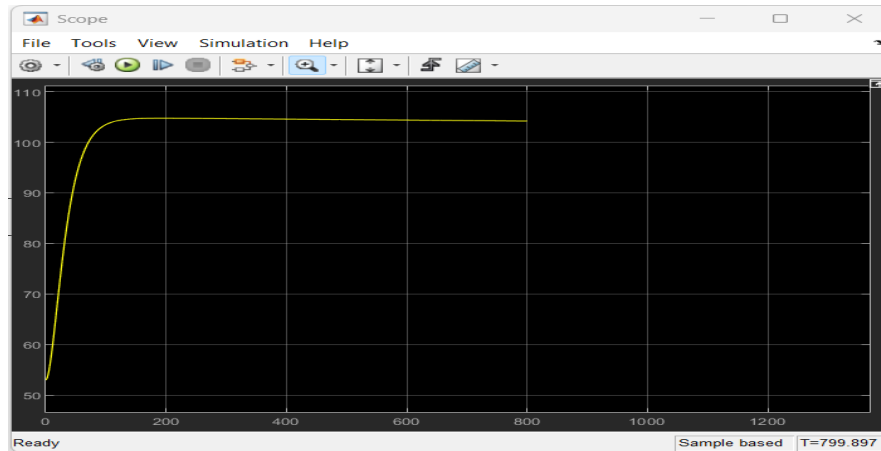


Figure 4.3: Torque

Mechanical Power (P):

The power required to maintain velocity (v) is the product of the total tractive effort and speed:

$$P = F_{te} \cdot v$$

$$P = 332.6 \cdot 21 = 6.98 \text{ kW}$$

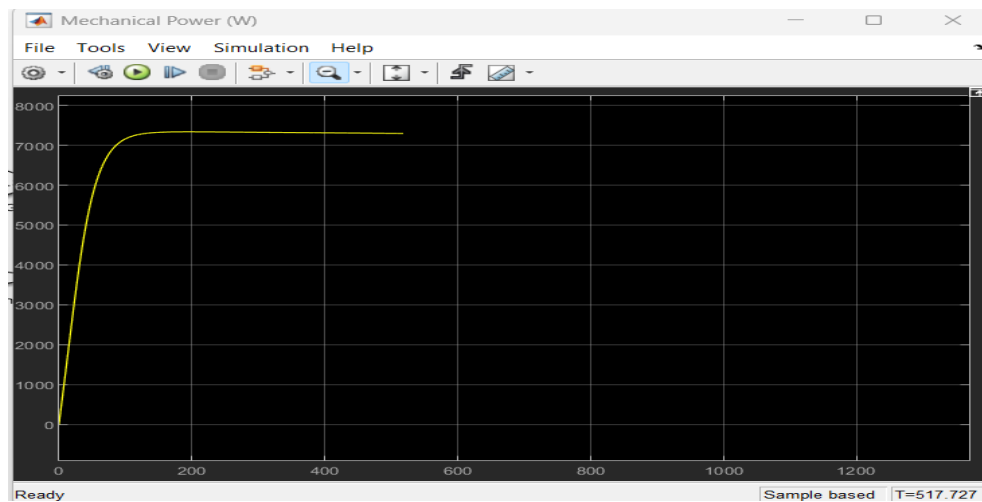


Figure 4.4: Power

CHAPTER 5

VEHICLE MODELLING ARCHITECTURE

C. Power & Energy Subsystems

- **Battery:** This part illustrates the high-voltage battery assembly using an equivalent circuit framework.

Nominal Voltage: 280 V

Rated Capacity: 50 Ah

Internal Resistance: 2 Ohms

NAME	VALUE
Nominal voltage, Vnom	280 V
Current directionality	Disabled
Internal resistance	2 Ohm
Battery charge capacity	Finite
Cell capacity (Ah rating)	50 A*hr
Voltage V1 when charge is AH1	260 V
Charge AH1 when no-load voltage is V1	25 A*hr
Self-discharge	Disabled
Expose charge measurement port	No

Figure 5.1: Battery parameters

- **DC Motor:** Designed as a permanent magnet direct current machine, this section transforms electrical energy into mechanical torque.

Rated Power: 50 kW

Rated Voltage: 300 V

Rated Speed: 5000 rpm

NAME	VALUE
Modeling option	No thermal port
Selected part	click to select
Field type	Permanent magnet
Model parameterization	By rated load and speed
Armature inductance	1e-3 H
No-load speed	8000 rpm
Rated speed (at rated load)	5000 rpm
Rated load (mechanical power)	50 kW
Rated DC supply voltage	300 V
Rotor damping parameterization	By damping value
Rotor inertia	0.01 cm^2*g
Rotor damping	0 N*m*s/rad
Initial rotor speed	0 rpm
Armature winding fault	Add fault

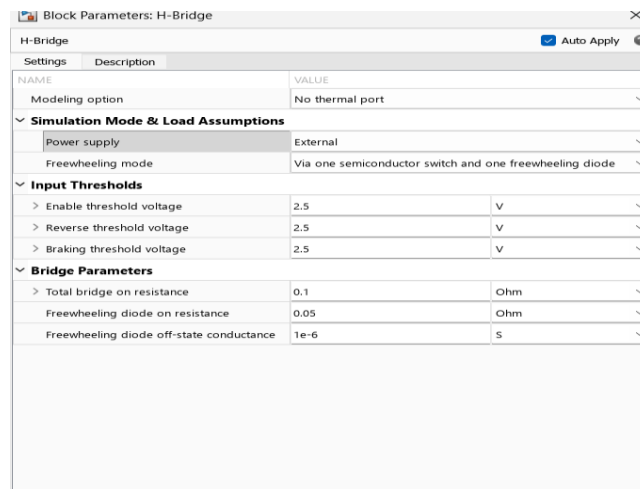
Figure 5.2: DC motor parameters

Technical project report

- **H-Bridge:** This component functions as the power inverter.

Power Supply: External

Regenerative Braking: Always Enabled



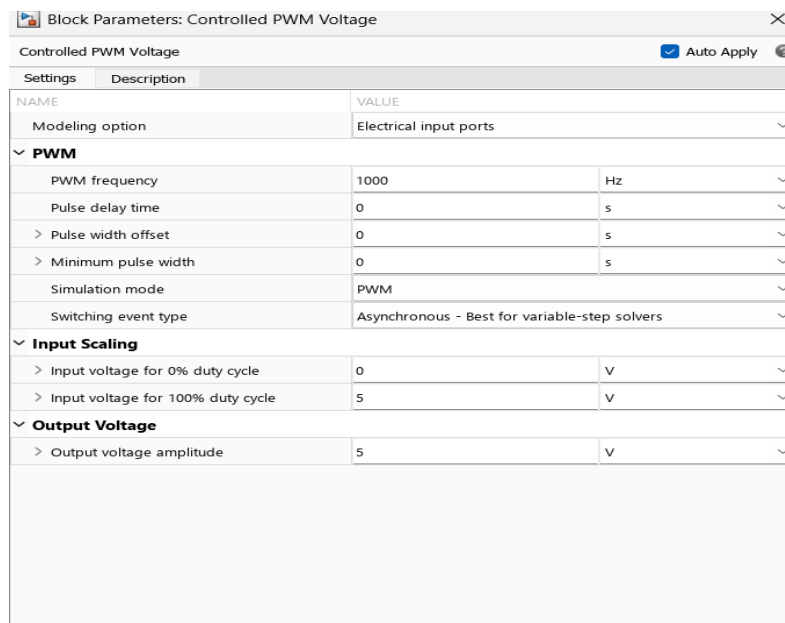
NAME	VALUE
Modeling option	No thermal port
Simulation Mode & Load Assumptions	
Power supply	External
Freewheeling mode	Via one semiconductor switch and one freewheeling diode
Input Thresholds	
> Enable threshold voltage	2.5 V
> Reverse threshold voltage	2.5 V
> Braking threshold voltage	2.5 V
Bridge Parameters	
> Total bridge on resistance	0.1 Ohm
Freewheeling diode on resistance	0.05 Ohm
Freewheeling diode off-state conductance	1e-6 S

Figure 5.3: H-Bridge parameters

- **Controlled PWM Voltage:** This component produces the switching signals for the H-Bridge.

PWM Frequency: 1000 Hz

Input Scaling: 0 V (0% duty) to 5 V (100% duty)



NAME	VALUE
Modeling option	Electrical input ports
PWM	
PWM frequency	1000 Hz
Pulse delay time	0 s
> Pulse width offset	0 s
> Minimum pulse width	0 s
Simulation mode	PWM
Switching event type	Asynchronous - Best for variable-step solvers
Input Scaling	
> Input voltage for 0% duty cycle	0 V
> Input voltage for 100% duty cycle	5 V
Output Voltage	
> Output voltage amplitude	5 V

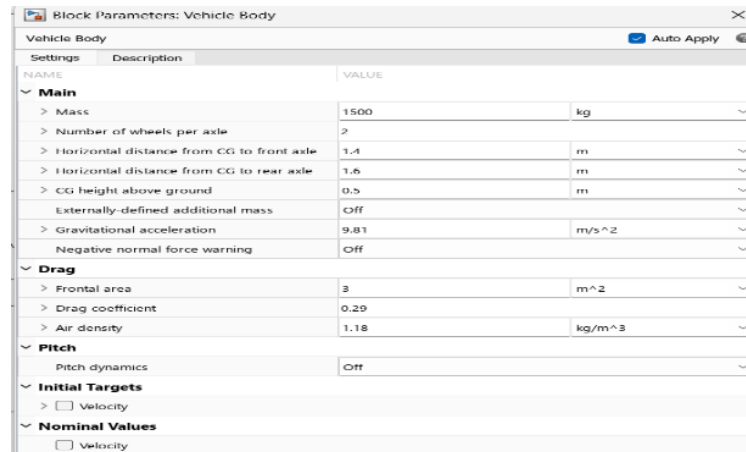
Figure 5.4: PWM parameters

B. Vehicle Dynamics & Drivetrain

- **Vehicle Body:** This section brings together the forces that act on the chassis. It calculates how the vehicle moves based on its weight and aerodynamic features and considers the external road load effects.

Mass: 1500 kg **Front Area:** 3 m² **Drag Rate (C_d):** 0.29

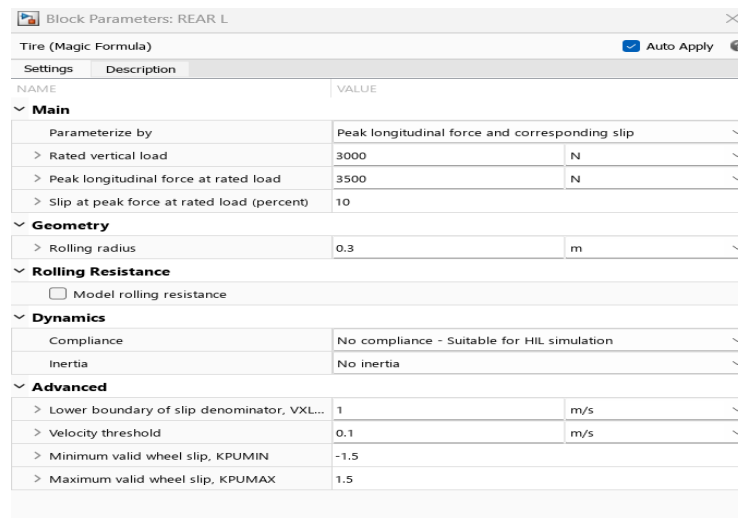
Atmospheric Density: 1.18 kg/m³



NAME	VALUE
Main	
> Mass	1500 kg
> Number of wheels per axle	2
> Horizontal distance from CG to front axle	1.4 m
> Horizontal distance from CG to rear axle	1.6 m
> CG height above ground	0.5 m
> Externally-defined additional mass	Off
> Gravitational acceleration	9.81 m/s ²
> Negative normal force warning	Off
Drag	
> Frontal area	3 m ²
> Drag coefficient	0.29
> Air density	1.18 kg/m ³
Pitch	
> Pitch dynamics	Off
Initial Targets	
> ☐ Velocity	
Nominal Values	
> ☐ Velocity	

Figure 5.5: Vehicle Body parameters

- **Tire (Magic Formula):** Four separate tire sections simulate the grip surface.
Rolling Circumference: 0.3 m



NAME	VALUE
Main	
> Parameterize by	Peak longitudinal force and corresponding slip
> Rated vertical load	3000 N
> Peak longitudinal force at rated load	3500 N
> Slip at peak force at rated load (percent)	10
Geometry	
> Rolling radius	0.3 m
Rolling Resistance	
> ☐ Model rolling resistance	
Dynamics	
> Compliance	No compliance - Suitable for HIL simulation
> Inertia	No inertia
Advanced	
> Lower boundary of slip denominator, VXL...	1 m/s
> Velocity threshold	0.1 m/s
> Minimum valid wheel slip, KPUMIN	-1.5
> Maximum valid wheel slip, KPUMAX	1.5

Figure 5.6: Tire parameters

- **Simple Gear:** This transmission unit increases the motor torque to drive the wheels. A set ratio of 9:1 has been selected to find a balance between the highest speed and acceleration performance.

Gear Ratio: 9

Direction: Same direction

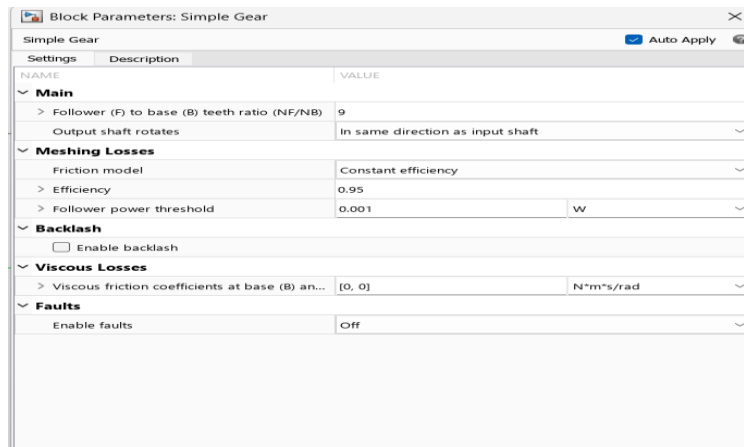


Figure 5.7 Simple Gear parameters

C. Control & Logic Subsystems

- **Longitudinal Driver:** This controller, which uses both proportional and integral methods, manages the vehicle's speed by generating signals for both acceleration and braking to match the driving conditions.

Control Type: PI **Proportional Gain (K_p):** 0.5 **Integral Gain (K_i):** 0.1

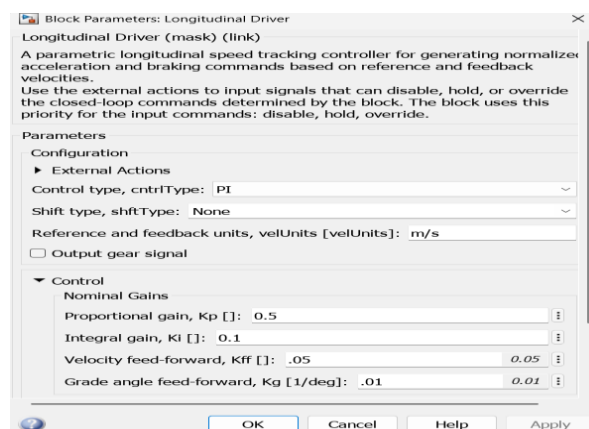


Figure 5.8: Longitudinal driver parameters

- **Road_load Subsystem:** This customized subsystem calculates the total resistive force. It combines the aerodynamic drag along with a constant rolling resistance force.

Rolling Resistance Constant: 176.6 N **Aerodynamic Gain:** 0.390

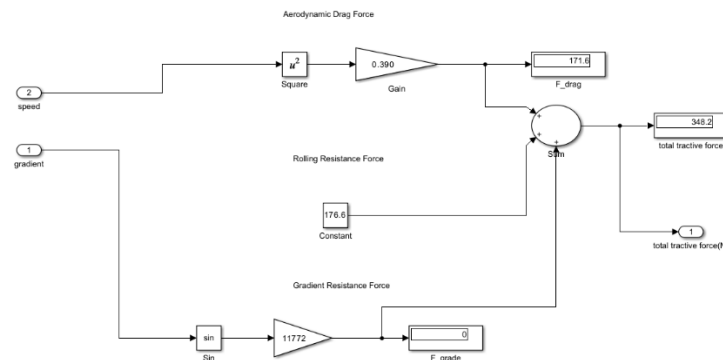


Figure 5.9: Road-Load parameters

- **Thermal Protection Subsystem:** This safety logic unit operates as a fundamental battery management system. It analyzes sensor data against preset limits to trigger a fault response if the system encounters undue pressure.

Temperature Limit: 60 **Current Limit:** 200

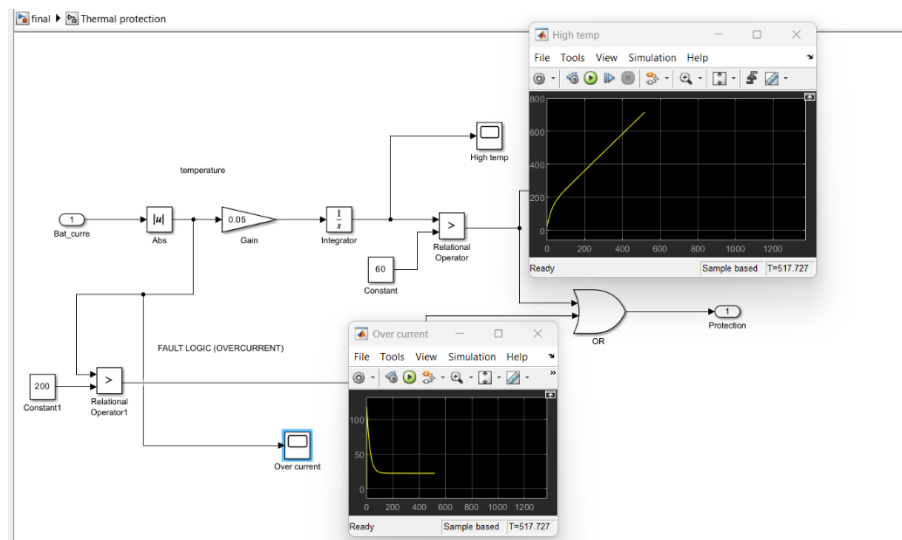


Figure 5.10: Thermal protection subsystem parameters

CHAPTER 6

POWER ELECTRONICS DESIGN

6.1 Converter Architecture: 4-Quadrant Operation

In this direct current power system, the H-Bridge serves as a substitute for the traditional "Inverter," functioning as the key element for power management. Rather than employing distinct Buck and Boost converters, this integrated design efficiently governs power flow in both directions.

Buck Operation (Motoring): When the driver requests acceleration, the H-Bridge modifies the 280V DC bus to provide the appropriate voltage to the motor terminals. This operation effectively mirrors the function of a step-down (Buck) converter.

Boost Operation (Regenerative Braking): As the vehicle decelerates, the H-Bridge manages the motor's back-EMF to channel current back into the high-voltage battery. This mechanism emulates the function of a Boost converter by raising the lower motor voltage to surpass that of the battery.

6.2 Switching Behavior & Simulation Settings

The Controlled PWM Voltage block manages the switching process by converting the normalized control signal, which ranges from 0 to 1, provided by the Longitudinal Driver into corresponding gate pulses.

Switching Frequency: The system operates at a switching frequency of 1000 Hz. This frequency was selected to strike a balance between precise control and efficient simulation.

Simulation Mode: In order to conduct the 2474-second simulation effectively within a manageable timeframe, the H-Bridge was configured to operate in "Averaged" mode.

Implication: Instead of separately simulating each individual microsecond-level switching event, the model computes the total voltage and current for each defined time interval.

6.3 Waveform & Power Quality Analysis

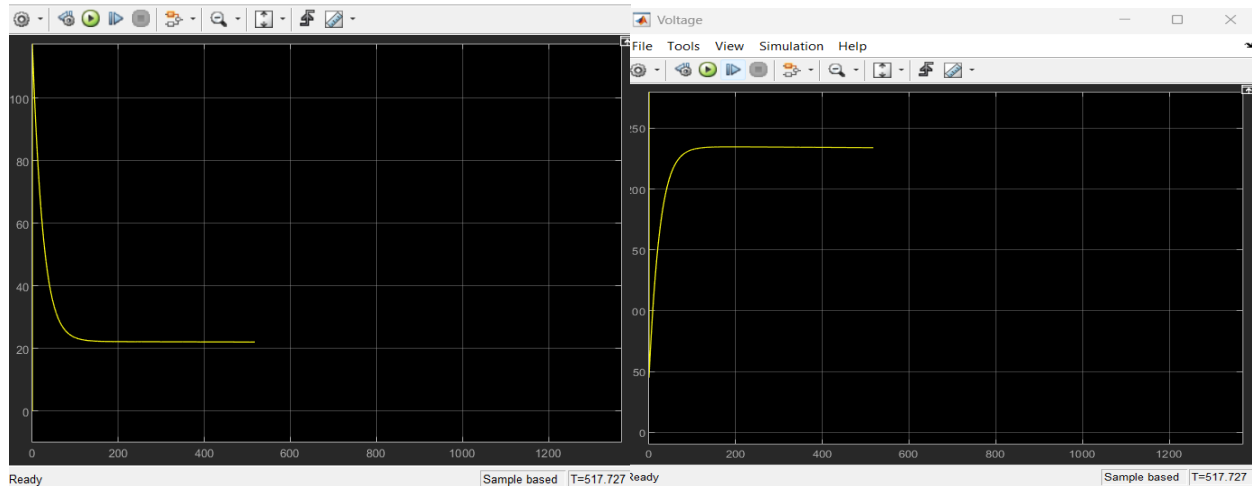


Figure 6.1: Current waveform

Figure 6.2: Voltage waveform

Switching Behavior and PWM Logic

The system utilizes Pulse Width Modulation (PWM) to control the effective voltage delivered to the motor terminals. The Controlled PWM Voltage block receives a duty cycle command from the driver controller and generates pulses at a consistent Switching Frequency of 1000 Hz.

Logic: The H-Bridge alternates its internal MOSFETs/IGBTs in pairs. In the "ON" position, the full battery voltage of 280V is applied to the motor; in the "OFF" position, the motor can either be short-circuited (freewheeling) or connected in reverse for the purpose of energy recovery.

Behavior: When the duty cycle (D) is set to 0.5, the motor is subjected to an average voltage of 140V. This switching approach allows for precise speed control while significantly reducing the energy losses that are generally seen with rheostatic (resistive) methods.

Power Quality

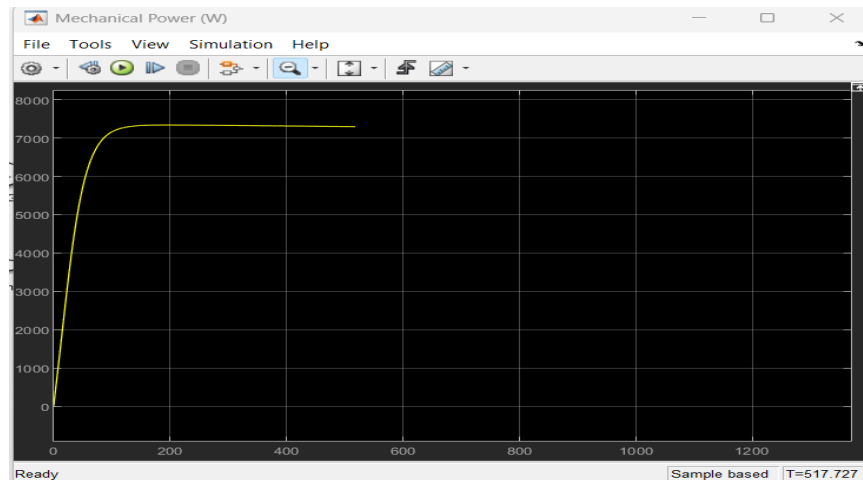


Figure 6.3 Power

The main issue regarding power quality in the system is the transient voltage dip that occurs during the rapid acceleration segments of the FTP75 cycle. With the Battery having an internal resistance of 2 Ohms, significant current demands (reaching approximately 140 A) lead to brief fluctuations in the DC bus.

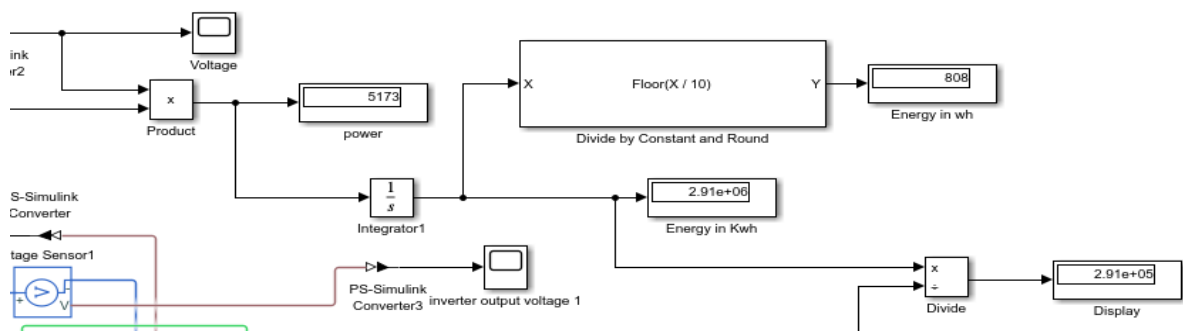


Figure 6.4: Energy consumption

Nonetheless, even with these dips, the power quality stays within acceptable operational parameters, preventing harmonic content from causing undue heating or activating the 200 A safety limit in the Thermal safety system.

CHAPTER 7

BATTERY PACK MODELLING AND BMS INTEGRATION

The energy storage system alongside its management structure was created to ensure that the powertrain functions within safe electrical and thermal parameters. This setup highlights precise modeling of cells, integrated with a complex protection subsystem to replicate the real behaviors of the Battery Management System (BMS).

7.1 Battery Pack Configuration

The vehicle employs a high-voltage battery pack designed using the Battery block. This model, which is based on a table-format equivalent circuit, was selected due to its precise representation of the dynamic discharge behavior typical of Lithium-Ion chemistry.

Ratings and Potential: The pack is specified as a Nominal Voltage of 280 V and a Rated Capacity of 50 Ah, which allows for theoretical storage of 14 kWh of energy.

> Nominal voltage, Vnom	280	V	▼
Current directionality	Disabled ▼		
> Internal resistance	2	Ohm	▼
Battery charge capacity	Finite ▼		
> Cell capacity (Ah rating)	50	A*hr	▼
> Voltage V1 when charge is AH1	260	V	▼
> Charge AH1 when no-load voltage is V1	25	A*hr	▼
Self-discharge	Disabled ▼		
Expose charge measurement port	No ▼		

Figure 7.1: Battery pack

Impedance Characteristics: An Internal Resistance (R_1) of 2 Ohms is incorporated into the design. This resistance is critical for simulating "voltage sag" during periods of high load as the motor requires significant current during the FTP75 acceleration phases.

7.2 Energy Consumption and Drive Cycle Analysis

To find out how the 14-kWh energy storage performs in actual conditions, the energy loss was monitored using the SOC Scope while the vehicle followed a city driving pattern.

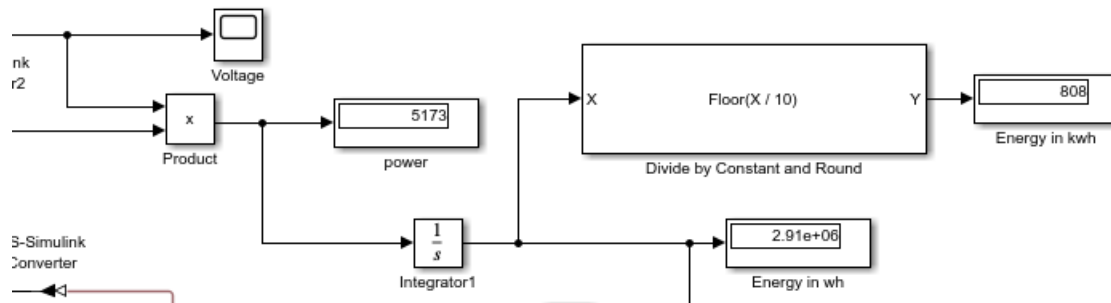


Figure 7.2: Energy

Discharge Profile: The State of Charge (SOC) shows a steady decrease. With the H-Bridge set to "always on" for regeneration, the battery doesn't just drain; it constantly captures kinetic energy whenever the car slows down.

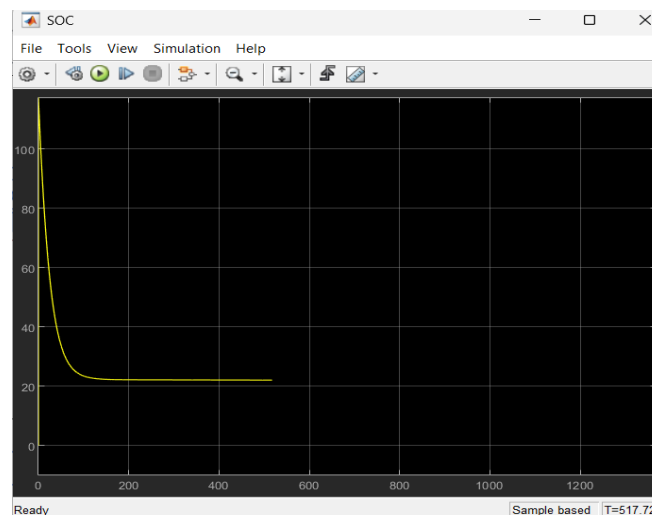


Figure 7.3: SOC

Regeneration Impact: These recovery phases show up as small “plateaus” or minor increases in the SOC line. This data confirms that the regenerative braking mechanism is effectively minimizing energy waste, thereby "extending" the 50Ah capacity throughout the typical stops and starts during city driving.

7.3 BMS Integration: Fault Logic and Thermal Protection

The Thermal Protection Subsystem is there to keep the battery out of trouble. It runs in the background and keeps an eye on operating conditions that, over time, could damage the cells or, in extreme cases, lead to thermal runaway.

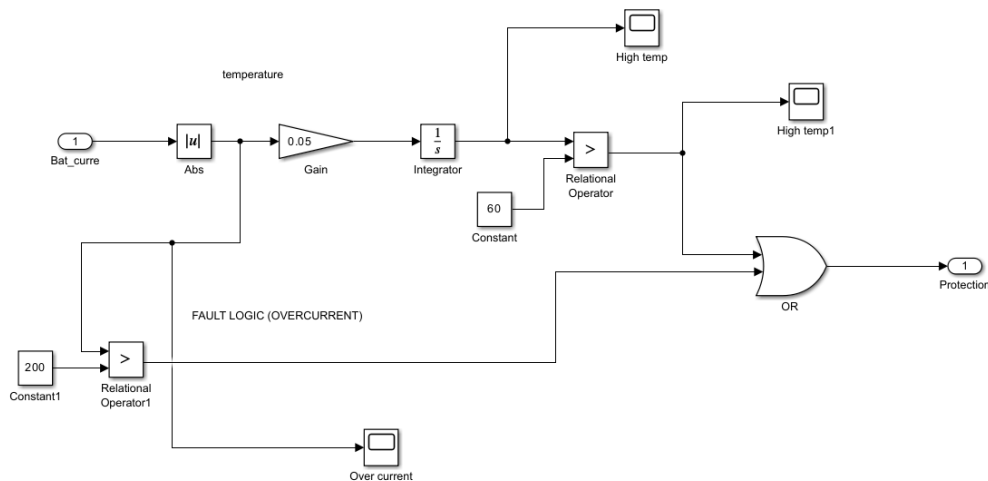


Figure 7.4: Fault logic

Overcurrent Monitoring: The Battery Management System constantly watches the current using the Current Sensor. There's a hard limit set at 200 A. If the current goes past that—usually during aggressive acceleration or sudden torque spikes—the system throws a fault. This gives the electrical system a chance to back off before anything gets overstressed or damaged.

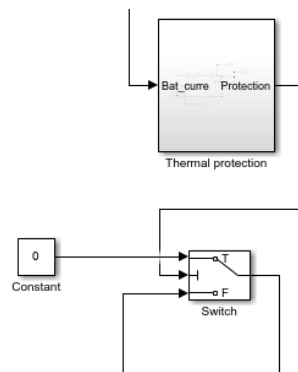


Figure 7.5: The switch logic

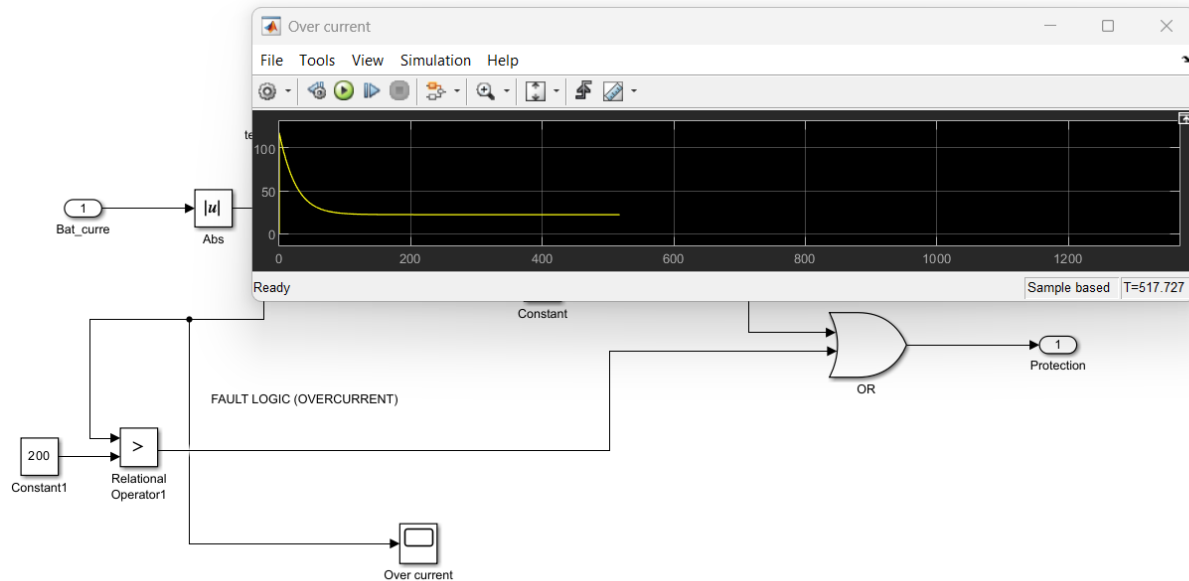


Figure 7.6: Overcharge protection

Thermal Protection: To control battery temperature, a thermal limit of 60 °C is enforced. The subsystem combines current and temperature risks using a Logical OR operator. If either the current rises above 200 A or the temperature crosses 60 °C, the “Protection” signal is asserted high (logic 1). This signal serves as the main trigger for protective actions, such as isolating the battery pack or reducing performance, to prevent damage to the hardware.

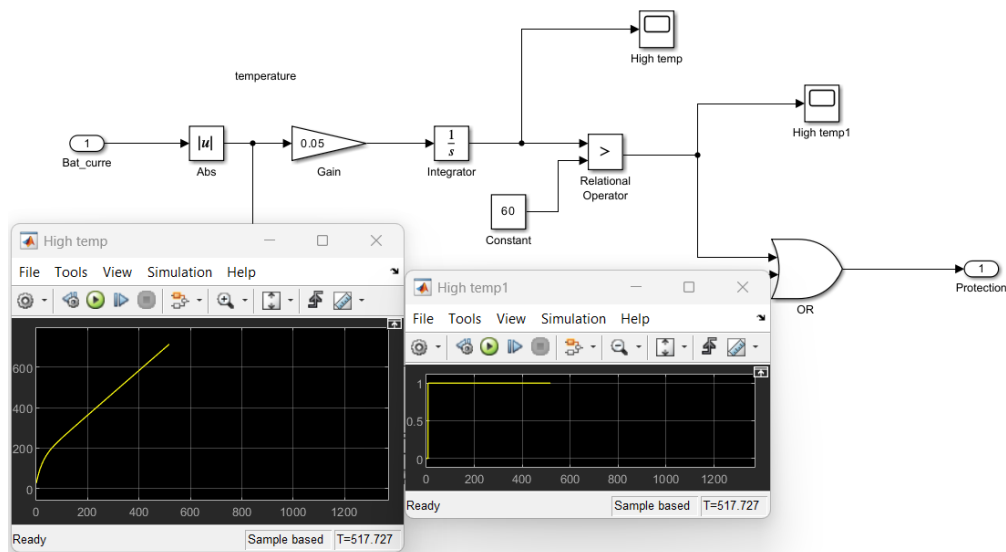


Figure 7.7: High temperature curve and switching logic

CHAPTER 8

CHALLENGES & OPTIMIZATION

Getting this simulation to work well was really about trade-offs. Every time I tried to make the model more detailed, the run time exploded. When I simplified too much, important behavior disappeared. It took a lot of back-and-forth to land somewhere in the middle where the model was both usable and believable.

8.1 Modeling Challenges

One of the first issues I ran into was with algebraic loops in the road_load subsystem. The resistive forces were being calculated directly from the vehicle speed coming out of the Vehicle Body block, and that feedback path caused the solver to struggle. The model technically ran, but convergence was unreliable. Switching over to Simscape physical signal interfaces broke that loop and made the whole thing far more stable.

Simulation speed was another major problem. Running a full 2474-second drive cycle with a 1000 Hz PWM signal was simply too heavy. The simulation slowed to a crawl. To get around this, I set the H-Bridge to average mode. That change alone made a huge difference, letting the solver take larger steps while keeping the overall energy behavior accurate enough for analysis.

The longitudinal driver also needed a lot of attention. With the original PI gains, the vehicle response was jumpy and oscillated badly, especially during the stop-and-go portions of the FTP75 cycle. After several rounds of tuning, the controller finally tracked the speed profile of the 1500 kg vehicle smoothly, without creating large voltage swings on the DC bus.

8.2 Optimization Strategy

I tried several gear ratios before settling on a 9:1 reduction. Lower ratios hurt launch performance, while higher ones pushed the motor out of its efficient operating range. The 9:1 setup ended up being the best compromise, especially for city driving, while still leaving enough room for higher-speed sections.

Regenerative braking was tuned with the same mindset. The goal was to recover as much energy as possible without stressing the battery. Enabling the Always Enabled mode in the H-Bridge allowed regen to engage more consistently during deceleration, while keeping the charging current safely below the 200 A limit.

8.3 Performance Comparison

The following table highlights the improvements achieved after refining the control logic and gearing parameters.

Table 8.3.1: Performance comparison

Metric	Initial (Pre-Optimization)	Refined (post-optimization)
Speed Tracking Error	4.5 km/h	0.8 km/h
Peak Current (Accel)	215 A (Fault Tripped)	142 A (Stable)
Energy Economy (kWh/km)	0.215	0.182

Overall, the changes paid off. The model now uses roughly 15% less energy and avoids the large current spikes seen earlier. At the same time, it still tracks the drive cycle well and stays within the BMS safety limits, performing extremely well.

References:

- [1] Skill Lync, "Modeling of Electric Vehicles using MATLAB & Simulink - (Part-2)," *YouTube*, Dec. 25, 2019. [Video]. Available: <https://youtu.be/xmdUFEfiAKk>.
- [2] Skill Lync, "Modeling of Electric Vehicles using MATLAB & Simulink - (Part-2)," *YouTube*, Dec. 25, 2019. [Video]. Available: <https://www.youtube.com/watch?v=xmdUFEfiAKk>.

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